

Retail Sales Performance Dashboard Documentation

I. Project Overview

A- What does the company sell?

The company sells retail products in different categories:

- Furniture
- Office Supplies
- Technology

The company sells products to different customer segments:

- Consumer
- Corporate
- Home Office

The sales happen in many cities and regions in the United States.

B- Goal of this project

The goal of this project is to analyze:

- Sales performance
- Profit performance
- Discount impact
- Return impact
- Shipping performance
- Orders

The dashboard helps management to:

- Make better business decisions
 - Find weak areas
 - Improve profit
 - Reduce returns
 - Control discounts
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2. Dataset

The dataset contains retail orders information.

The dataset also contains:

- Shipping information (Ship Date – ship type)
- Product information (Category- Sub-Category- Product Name)
- Customer information (Customer Name- Segment)
- Financial information (Sales- Profit- Discount- Quantity)
- Country information (State - City- Region)

Others columns:

- Order ID
- Order Date
- Salesperson
- Returned Status

3. Data Preparation

A- General Data Cleaning: -

The following steps were taken:

- Promoted Headers
- Remove duplicates
- Remove other columns
- Remove blanks
- Replace errors
- Changed data types

B- Enhancement

New calculated columns were created:

1- Cost Column

Cost = Sales - Profit

Used to calculate total cost.

2- Sales before Discount

Sales before discount = Sales / (1 - Discount)

Used to know original sales before discount.

3- Amount Discount

Amount discount = sales before discount - Sales

Used to calculate discount value.

4- Profit before Discount

Profit before discount = sales before discount - cost

Used to estimate profit before discount.

5- Days to Ship

DATEDIFF (Order Date, Ship Date, DAY)

Used to measure shipping speed.

4. Data Modeling

The model contains:

- Fact table for sales data
- Calendar table
- Supporting tables for waterfall charts
- Parameter table

Relationships were created between:

- Order Date and Calendar Date

The model uses:

- One-to-many relationships

5. Calculations (DAX)

1. Orders

☞ Goal: track number of orders

KPIs:

- Total Orders
 - ☞ This shows how many orders we have
- Final Total Orders
 - ☞ This shows orders after removing returned orders
- Average Order Value (AOV)
 - ☞ average sales per order
- Avg Basket Size
 - ☞ average quantity per order

Measures used:

- `DISTINCTCOUNT (Order ID)`
 - ☞ because one order can contain many products in different rows
- `CALCULATE ()`
 - ☞ used to remove returned orders from calculation

2. Sales

☞ Goal: track company revenue performance

KPIs:

- Sales Before Discount
 - ☞ total sales before applying discount
- Sales After Discount
 - ☞ sales after discount
- Final Sales
 - ☞ final sales after removing returned orders

Measures used:

- `SUM (Sales)`
 - ☞ used to calculate total sales

- `CALCULATE ()`
☞ used to exclude returned orders
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3. Profit

☞ Goal: measure business profitability

KPIs:

- Profit Before Discount
☞ estimated profit before discount
- Total Profit After Discount
☞ actual profit after discount
- Final Profit
☞ profit after removing returned orders
- Profit Margin
☞ shows percentage of profit from sales

Measures used:

- `SUM(Profit)`
☞ calculate total profit
- `DIVIDE ()`
☞ used for safe division and avoid errors

4. Cost

☞ Goal: calculate product cost

KPIs:

- Total Cost
☞ total product cost
- Final Total Cost
☞ total cost after excluding returned products

Measures used:

- `Sales - Profit`
☞ cost calculated from sales minus profit
 - `SUM()`
☞ calculate total cost
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5. Discount

☞ Goal: analyze discount effect on business

KPIs:

- Amount Discount
☞ total discount value
- Discount Rate
☞ percentage of discount from total sales

Measures used:

- `DIVIDE ()`
☞ calculate discount percentage
 - `Sales / (1 - Discount)`
☞ calculate sales before discount
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6. Return Analysis

☞ Goal: measure impact of returned orders

KPIs:

- Return Rate
 - ☞ percentage of returned orders
- Total Returns
 - ☞ total sales value lost returns

Measures used:

- `CALCULATE ()`
 - ☞ filter returned orders only
 - `DIVIDE ()`
 - ☞ calculate percentages safely
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7. Customer Analysis

☞ Goal: understand customer behavior

KPIs:

- Total Customers
 - ☞ total unique customers

Measures used:

- `DISTINCTCOUNT (Customer ID)`
 - ☞ count unique customers
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8. Shipping Analysis

☞ Goal: measure shipping performance

KPIs:

- Average Days of Ships
 - ☞ average delivery time between order and shipping

Measures used:

- `DATEDIFF ()`
 - ☞ calculate shipping days
 - `AVERAGE ()`
 - ☞ calculate average shipping time
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9. Time Intelligence

☞ Goal: compare current performance with previous periods and track growth over time.

KPIs:

- Profit Last Year (Profit LY) → shows profit from the previous year
- Profit Growth % → shows increase or decrease compared with last year

Measures used:

- `DATEADD ()` → used to shift the date by one year back
- `CALCULATE ()` → used to calculate values in a different time period
- `DIVIDE ()` → used to calculate growth percentage safely

10. Waterfall Analysis

☞ Goal: explain sales movement step by step

KPIs:

- Waterfall Value2
☞ shows relationship between sales, returns, cost, and profit
- Waterfall Value
☞ shows movement from sales before discount to net sales

Measures used:

- SWITCH()
☞ change values based on waterfall step
- Negative values -Measure
☞ used to show decrease in waterfall chart

6. Dashboard Layout

General Flow

The dashboard is designed to help users understand business performance step by step. The user starts from general sales performance, then moves to discount analysis, return analysis, shipping performance, Orders and finally business insights.

The dashboard uses:

- KPIs
- Charts
- Filters
- Dynamic measures
- Interactive visuals

to make data easy to understand and support decision making.

Page 1 – Finance Dashboard

☞ Goal: provide a complete financial overview of business performance

This page includes:

- Total Sales
- Total Profit
- Profit Margin
- Total Customers
- Total Orders
 - Discount Rate
 - Return Amount
 - Total Cost

Visuals Used:

- KPI Cards
☞ to display main business indicators quickly
- Line Chart
☞ to show Margin Profit trend over years

- **Bar Charts**

☞ to compare:

- Segment performance
- Region performance
- Category profitability
- Retail salespeople's performance

- **Waterfall Chart**

☞ to explain how sales move to final profit through:

- Cost
- Returns
- Profit

- **Matrix Tables**

☞ to analyze yearly sales growth and profit growth

➤ In this page, a parameter is added to switch between (Sales, Profit and margin profit)

Page 2: Returned Analysis Dashboard

Goal:

Analyze returned orders and understand their impact on business performance.

Visuals Used:

KPI Cards

Show sales, profit, margin profit, return rate, returned quantity, and returned orders.

Line Chart (Returns value and # Returns orders by Year)

Shows return trend across years.

Column Chart (Returns value and # Returns orders by region)

Shows return performance across regions.

Donut Chart (Returns value and # Returns orders by Category)

Shows return distribution among categories.

Bar Chart (Returns value and # Returns orders by Sub-Category)

Shows sub-categories with the highest returned value.

Donut Chart (Returns value and # Returns orders by Segment)

Shows return contribution by customer segments.

Bar Chart (Returns value and # Returns orders by State)

Shows states with highest returns.

Column Chart (# Returns orders and average days of ships by Ship Mode)

Shows relationship between shipping methods and returned orders.

Page 3: Orders Analysis Dashboard

Goal:

Analyze order volume and customer purchasing behavior.

Visuals Used:

KPI Cards

Show sales, profit, margin profit, cost, total orders, AOV, and basket size.

Line Chart (# Orders by Year)

Shows order growth over years.

Donut Chart (# Orders by Segment)

Shows order distribution across customer segments.

Bar Chart (# Orders by City)

Shows cities with highest order counts.

Column Chart (# Orders by Ship Mode)

Shows order distribution among shipping methods.

Column Chart (# Orders by Region)

Shows regional order performance.

Column Chart (# Orders by Retail Salespeople)

Shows salespeople order performance.

Page 4: Shipping Analysis Dashboard

Goal:

Evaluate shipping efficiency and delivery performance.

Visuals Used:

KPI Cards

Show sales, profit, margin profit, cost, returned orders, and average shipping days.

Line Chart (average days of ships by Year)

Shows shipping trend over time.

Pie Chart (average days of ships by Segment)

Shows shipping performance by customer segment.

Bar Chart (average days of ships by City)

Shows cities with highest shipping duration.

Column Chart (average days of ships and # Returns orders by Ship Mode)

Shows shipping performance across shipping methods.

Column Chart (average days of ships by region)

Shows shipping duration differences across regions.

Page 5: Discount Analysis Dashboard

Goal:

Analyze discount strategy and evaluate its impact on sales performance and profitability.

Visuals Used:

KPI Cards

Show sales, profit, margin profit, cost, discount rate, and discount value.

Line Chart (Discount Rate by Year)

Shows discount trend across years.

Bar Chart (Discount Rate by State)

Shows states with highest discount rates.

Column Chart (Discount Rate by Retail Salespeople)

Shows discount rates applied by salespeople.

Column Chart (Discount Rate by Category)

Shows discount distribution across categories.

Waterfall Chart (Waterfall Value by Step)

Shows sales movement from sales before discount to net sales.

Column Chart (Discount Rate by Region)

Shows discount differences across regions.

Column Chart (Discount Rate by Segment)

Shows discount distribution across customer segments.

Bar Chart (Discount Rate by Sub-Category)

Shows sub-categories with highest discount rates.

7. Interaction

Goal:

Enhance user experience by allowing dynamic exploration and filtering of data across all dashboards.

Interaction Features:

Slicers (Filters)

Used to filter data by Year, Region.

Cross Filtering (Cross Highlighting)

Selecting any visual automatically updates and highlights related data in other visuals to show relationships.

Drill Down / Drill Up

Allows users to explore data from high-level views (Year/Region) down to detailed levels (Month/State/City).

Tooltips (Insight on Hover)

Displays additional insights when hovering over charts, providing quick explanations without leaving the page.

Navigation Buttons

Enable smooth movement between dashboard pages (Finance, Returns, Orders, Shipping, Discount).

Dynamic Titles

Page titles and KPIs change based on selected filters (e.g., selected stage or segment).

Waterfall Interaction

Allows step-by-step analysis of financial changes (Sales → Discount → Returns → Net Sales).

Drill Through Pages

Allows users to navigate from summary information to detailed analysis pages.

- **City Drill Through** and **Product Drill Through**
- ☞ This helps users move from high-level insights to detailed investigation quickly.

8. Conclusion

This project helps understand business performance by analyzing sales, profit, orders, returns, shipping, and discounts. The dashboard shows that some regions and categories perform better than others. It also shows that high discounts and returns reduce profitability. Interactive features such as filters, drill through, and tooltips help users find insights quickly and support better business decisions.